Taqi Haider

taqihaider592@gmail.com Taqi-Haider
http://taqihaider7.github.io/

Education

Sep 2024 – Jul 2026 **MS, Artificial Intelligence and Big Data Analytics**, Novosibirsk State University, Russia.
Thesis: *Medical Dataset Name Extraction from Scientific Articles*.
GPA: **4.9/5.0**; graduated with honours (Red Diploma).

Dec 2016 – Dec 2020 **BE, Electrical Engineering**, Quaid-e-Awam University of Engineering, Science and Technology, Pakistan.
Thesis: *A Comparative Study on PWM Techniques of an AC Voltage Controller in Terms of Harmonic Content*.
CGPA: **3.65/4.0**.

Research Interest

- **Multimodal Large Language Model**
- **Interpretability & Controllability of Large Language Models**
- **Natural Language Processing**

Research Experience

Nov 2024 – Present **AI Researcher**, Stream Data Analytics and Machine Learning Lab, Novosibirsk State University.

- Develop and evaluate methods for extracting structured information from scientific articles.
- Coordinate master's students' industrial-practice research projects.
- Contribute to AI-agent integration and dataset curation in the Kappa Framework.

Selected Research Projects

2024 – 2026 **MedSETNER: medical dataset-name extraction** (MS thesis; research lead).

- Led corpus development and annotation across **691 articles** containing **9,376 dataset-name mentions**; inter-annotator agreement was Fleiss' $\kappa = 0.85$.
- Compared rules, CRF, SciBERT, ModernBERT, BiomedBERT and GLiNER2 using article-level splits and a separate 151-article out-of-distribution test set. BiomedBERT achieved mean exact-span F1 of **60.7%** (same-pool) and **60.8%** (out-of-distribution).

2025 – 2026 **ClaimGraph-BT: claims in brain-tumour AI literature** (model training and evaluation).

- Trained and evaluated context-enhanced biomedical claim detectors using paper-level splits and manually verified test data; PubMedBERT Context achieved **0.8391 macro F1** and **0.9301 AUC**.
- Evaluated structured claim extraction and deterministic repair; macro field-presence F1 increased from **0.1668** to **0.4850** after repair and grounding.

Publications and Scholarly Outputs

Accepted conference papers

2026 **Taqi Haider**, Nawazish Hussain, Syed Muhammad Irtiza Zaidi, and Evgeniy Nikolaevich Pavlovskiy. "MedSETNER: A Benchmark Corpus for Extracting Dataset Names from Medical Scientific Literature." Accepted at the *13th International Conference on Analysis of Images, Social Networks and Texts (AIST 2026)*, CCIS track.

Nawazish Hussain, **Taqi Haider**, and Evgeniy Nikolaevich Pavlovskiy. "ClaimGraph-BT: Automated Claim Extraction and Cross-Paper Comparison for Brain Tumor AI Literature." Accepted at *AIST 2026*, CCIS track.

Publications and Scholarly Outputs (continued)

Dataset and conference abstract

- Nawazish Hussain and **Taqi Haider**. "BrainTumorClaims: A Dataset for Scientific Claim Detection and Structured Information Extraction in Brain Tumor Literature." Dataset, Novosibirsk State University. doi:10.5281/zenodo.20364566.
- **Taqi Haider**. "Automatic Extraction of Medical Dataset Names from Research Papers." Conference abstract accepted at BGRS/SB 2026.

Manuscript under review and ongoing work

- Dania Batool, **Taqi Haider**, Malik Ali Judge, Aisha Kanwal, and Jackie Lohana. "Artificial Intelligence for Plant Disease Detection: A Systematic Literature Review of Conventional and Advanced Methods." Manuscript under review at *Artificial Intelligence in Agriculture*.
- **Taqi Haider**, Anna Latushko, and Julia Mazine. "CivilNER: Task, Method and Dataset Mention Extraction from Civil Engineering Research Articles." Work in progress.

Teaching and Research Communication

Oct 2022 – Oct 2024 — **Data Science Instructor**, Skills Camp. Taught online machine-learning and web-scraping sessions and produced Urdu/Hindi instructional videos for the Skills Camp channel.

2023 — **Workshop speaker**, "Data & Delight: Pie & AI," an interactive analytics workshop organised by DeepLearning.AI.

2022 — **Invited speaker**, "Data Science for Agriculture," University of Agriculture, Faisalabad.

Skills

Languages	Strong reading, writing, and speaking competencies in English, Urdu, Sindhi, and Punjabi.
Coding	Python, R, TensorFlow, PyTorch, SQL, LaTeX, Linux,
Databases	MySQL, Vector Databases like Pinecone, Chroma, Qdrant, etc.
Cloud Computing	AWS, GCP, Azure.
Data Analysis/ Machine Learning	Academic Research, Data Analysis, Machine Learning, Deep Learning, Natural Language Processing, and Computer Vision.

Professional & Academic Experience

Academic & Volunteer Activities

2023 — **Data Science and Machine Learning Instructor, Content Creator** at Skills Camp YouTube Channel.

Speaker at Data & Delight: Pie & AI Interactive Analytics workshop organized by DeepLearning.AI

2022 — **Data Science for Agriculture Keynote Speaker** at University of Agriculture, Faisalabad.

2020 — **Director Simulation Projects**, PLUGIN-Project Expo, QUEST Nawabshah.

Awards & Achievements

2023 — **Got 1st Position in Cohere Hackathon** organized by lablab.ai

1000 Contribution Points to Aya Open Science Multilingual Large Language Model Project by Cohere for AI

Participated and Submitted solutions in OpenAI Hackathon organized by lablab.ai

2022 — **Won a Prize in Datadanics-1 Datathon** organized by Codanics.

Got Silver Medal in a Data Science competition organized by Kaggle

2019 — **Got a Laptop under Prime Minister's Laptop Scheme**

Professional & Academic Experience (continued)

MOOCs & Certifications

2023 **Machine Learning Specialization** by Deeplearning.AI, Stanford University

2022 **Data Science with Python** by Codanics

Data Analysis with Python by IBM

References

Dr. Maryam Asim
Lecturer
Department of Statistics, Quaid-i-Azam University, Islamabad, Pakistan,
Email: masim@qau.edu.pk

Dr. Evgeniy Pavlovskiy
Associate Professor
Faculty of Mechnaics & Mathematics, Novosibirsk State University, Russia
Email: e.pavlovskiy@g.nsu.ru